Store Entry/Exit People Counting

One-page approach note | YOLO26m + ByteTrack | Fixed 1270x720 CCTV camera

Approach

People are detected with the open-source **YOLO26m** model, restricted to the COCO *person* class. **ByteTrack** maintains a per-person ID across frames. The doorway gate is calibrated at the lower-right physical threshold from **(900,720) to (1260,500)**. The bottom-center foot point of each box is used for crossing decisions.

To reduce false events, the counter uses a line hysteresis band, stable-side dwell across multiple frames, a two-second initial-occupancy warm-up, and a per-track cooldown. Each accepted crossing updates the live headcount and is written to **timestamp, person_id, event, headcount**. The video overlay shows boxes, IDs, the gate, customer/staff labels, and live totals.

Edge-case handling

- **Occlusion:** ByteTrack bridges short detector gaps using motion and association. Hysteresis and stable-side dwell prevent a partially hidden box from immediately flipping sides. Long, complete occlusions can still cause an ID switch.
- **Groups entering together:** every visible person receives an independent detection and track, so side-by-side people generate separate events. Full body overlap can still merge detections and cause undercounting until people separate.
- **Re-entry:** state is tracked per ID and cooldown suppresses rapid jitter. A genuine OUT followed later by IN is accepted, even when a long disappearance causes a new ID.
- **Staff vs. customers:** a persistent cashier-zone polygon (**0,0;210,0;210,500;0,500**) marks tracks observed there for at least 0.75 seconds and 35% of visible frames as **STAFF**. Staff are orange and excluded from customer totals; all other tracks are green **CUSTOMER** tracks. This is a zone/behavior heuristic, not clothing recognition.

What failed / limitations

The wide-angle camera and heavy overlap near the entrance make perfect identity continuity impossible. A line counter cannot recover a person fully hidden behind a group, and the staff zone can misclassify a customer who enters the cashier area. The clip has no ground-truth annotations, so no benchmark accuracy is claimed; the CSV is an auditable event output.

What I would do with more time

> Annotate entrance-heavy validation segments; tune confidence, gate placement, and tracking thresholds; compare ByteTrack with BoT-SORT/ReID; add entrance-region tracklet stitching; fine-tune a camera-specific overhead-person detector; and replace the staff heuristic with a staff roster, calibrated zone policy, or trained appearance/ReID classifier. I would also export low-confidence/overlap cases for human review.